🎓 The Syllabus Sync

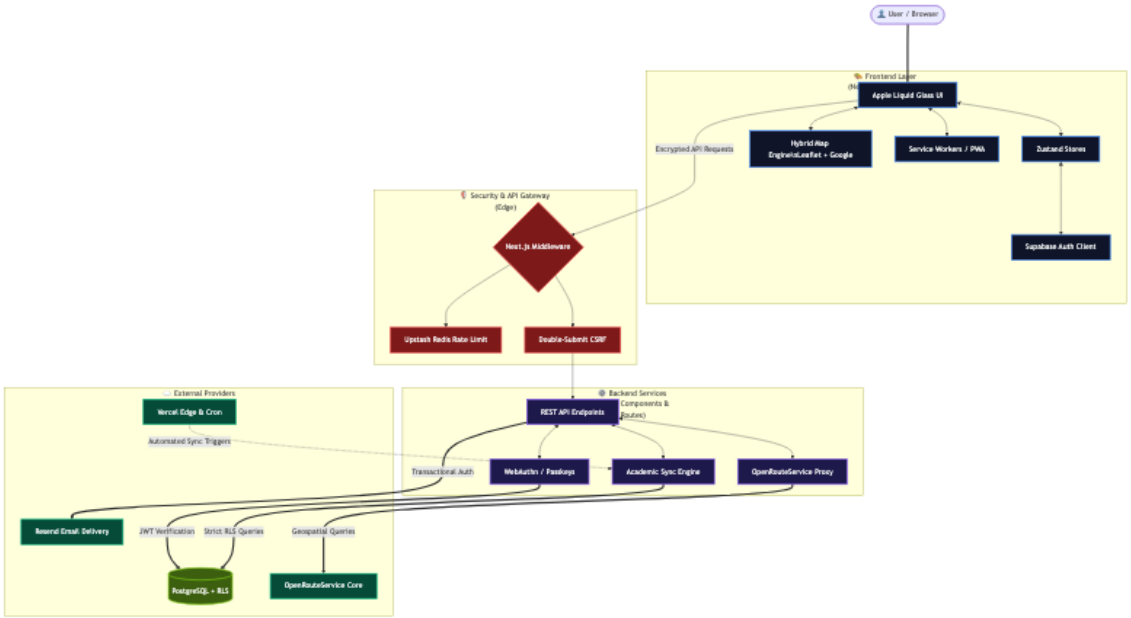
Enterprise-Grade Campus Management & Academic Productivity Platform *Next-Generation Student Experience Platform for Macquarie University*

🌟 Executive Summary

The Syllabus Sync is a comprehensive campus management ecosystem engineered specifically for Macquarie University students. Built on **Next.js 16** and **React 19**, the platform combines modern web technologies with thoughtful user experience design to deliver a high-performance productivity suite.

From precision campus navigation with real-time wayfinding to intelligent deadline tracking, Syllabus Sync represents the pinnacle of modern student experience design.

🗺️ System Architecture



🚀 Core Platform Features

📅 Academic Management System

- **Intelligent Calendar Engine:** Interactive weekly/daily views with real-time academic workload analysis.
- **Deadline Intelligence:** Smart prioritization with automated reminders and dependency tracking.

🗺️ Campus Navigation Platform

- **Precision Wayfinding System:** Dual-engine architecture featuring advanced Leaflet-based rendering and Google Maps integration.
- **Real-Time GPS Engine:** High-frequency tracker utilizing sensor-fusion and spatial threshold debouncing (~20m) to provide stable turn-by-turn navigation.

🎮 Academic Gamification Framework

- **Progression System:** XP earning through academic achievement and campus engagement.
- **Social Features:** Leaderboards, achievement sharing, and study group formation.

🔐 Security & Authentication

- **Multi-Factor Authentication:** Support for TOTP authenticators and FIDO2 Passkeys.
- **Custom Email Verification:** Production-ready verification via Resend with SHA-256 token hashing and rate limiting.

🛠️ Technology Architecture

Layer	Technologies
Frontend	React 19, Next.js 16 (Turbopack), Zustand, Framer Motion
Backend	Supabase (PostgreSQL + RLS), Node.js API Routes
Design	Tailwind CSS, Radix UI Primitives, Custom MQ Design Tokens
Security	Upstash Redis (Rate Limiting), CSRF Protection, CSP
QA	Vitest (Unit/Integration), Playwright, GitHub Actions CI/CD

🔒 Security & Quality Assurance

- **Multi-Layer Defense:** Distributed rate limiting, CSRF protection with double-submit cookies, and Content Security Policy (CSP).
- **Privacy-First Design:** GDPR & Australian Privacy Principles compliant, transparent data handling.
- **Comprehensive Testing:** Over 480 verified automated tests across unit, integration, and security layers.
- **Global Inclusivity:** Support for 19 languages alongside full accessibility compliance (WCAG 2.1 AA).

👥 Core Development Team

Leadership

- **Pouya Alavi** - *Frontend Architect* (UI/UX Design, Component Architecture, State Management)
- **Roauf Abedini** - *Backend Engineer* (API Development, Database Design, Security & Infrastructure)

Built with modern engineering practices by the Macquarie University community development team.